

Microstructural asymmetries of the planum temporale predict functional lateralization of auditory-language processing

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Abstract

Structural hemispheric asymmetry has long been assumed to guide functional asymmetry of the human brain, but empirical evidence for this compelling hypothesis remains scarce. Recently, it has been suggested that microstructural asymmetries may be more relevant to functional asymmetries than macrostructural asymmetries. To investigate the link between microstructure and function, we analyzed multimodal MRI data in 907 participants. We quantified structural and functional asymmetries of the planum temporale (PT), a cortical area crucial for auditory-language processing. We found associations of functional PT asymmetries and several microstructural asymmetries, such as intracortical myelin content, neurite density, and neurite orientation dispersion. The PT microstructure per se also showed hemispheric-specific coupling with PT functional activity. All these functional-structural associations are highly specific to within-PT functional activity during auditory-language processing. These results suggest that structural asymmetry guides functional lateralization of the same brain area and highlight a critical role of microstructural PT asymmetries in auditory-language processing.

eLife assessment

The authors studied the relationship between structural and functional lateralization in the planum temporale region of the brain, whilst also considering the morphological presentation of a single or duplicated Heschl's gyrus. The analyses are **convincing** due to a large sample size, inter-rater reliability, and corrections for multiple comparisons. The associations in this **valuable** work might serve as a reference for future targeted-studies on brain lateralization.

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Introduction

Whether and how neuroanatomy underpins hemispheric functional lateralization is a fundamental issue in neuroscience¹. One compelling notion is that “functional lateralization is guided by structural asymmetry”, hypothesizing a critical role of structural asymmetries in the neuroanatomical basis of functional lateralization^{2,3}. According to this hypothesis, a coupling of functional and structural asymmetries from the same brain area is expected. Notably, while both group-level functional and structural asymmetries have been reported for the same areas^{4–9}, such coexistence of the two asymmetries across studies did not represent a direct coupling between them. To rigorously determine the coupling of the two asymmetries in the same area, it is indispensable to empirically evaluate their association at the individual level.

The planum temporale (PT) is a well-known cortical area posterior to Heschl’s gyrus (HG, the auditory hub) and putatively plays an essential role in the functional lateralization of language processing^{8,10–16}. Hemispheric PT asymmetry in functional activation during auditory- or language-related tasks has been well demonstrated^{6,10,17–19}. Additionally, the PT shows distinct leftward asymmetries in a variety of structural aspects, e.g., length, gray matter concentration/volume, surface area, cortical thickness, and columnar neuronal units, although the results might conflict between studies^{14,20–33}. Due to the role of the PT in language processing, the idea that these asymmetries may be related to leftward functional asymmetries in speech processing is highly intuitive. Unexpectedly, examining the functional association of PT macrostructural asymmetries at the individual level almost always showed negative results, except for only a couple of studies demonstrating correlations between PT asymmetry of surface area and some non-PT-related indices of language lateralization, e.g., the asymmetry of dichotic listening^{34–36}.

Several factors possibly account for such an unexpected majority of negative results. First, the sample size in previous studies is too small for such association analyses across individuals, compared to the recently suggested thousands of individuals for capturing a reproducible brain-wide association³⁷. Next, structural asymmetries were confined to PT macrostructural measures that are biologically nonspecific, e.g., surface area and thickness. Recent studies have used novel MRI techniques to quantify PT microstructure on the level of axons or dendrites and further revealed PT asymmetries in myelin content, neurite density, and neurite orientation dispersion^{25,30–32}. Importantly, postmortem work on the microstructure of the temporal cortex has led to the hypothesis that asymmetries in the organization of the intrinsic microcircuitry of the PT and other areas may be crucial for functional lateralization²⁶. In line with this idea, asymmetries in PT microstructure have been linked to electrophysiological correlates of auditory speech, an indirect measure of functional language lateralization³⁰. However, the crucial experiment would be to show a direct link between PT microstructure and PT functional lateralization measured with fMRI. Finally, duplicated HG (dHG) in the left or right hemisphere occurs substantially in healthy individuals, although less frequently than single HG (sHG)^{38–40}. In particular, such an HG gyrification pattern is accompanied by significant changes in PT structural properties as well as their asymmetries compared with the single HG⁴¹. The interindividual variation in the HG gyrification pattern therefore should be carefully considered when investigating PT asymmetries as well as their functional-structural association, but this has been largely overlooked in previous studies.

In the present study, we included a large cohort of 907 healthy young adults to assess whether and how PT asymmetries of speech-related functional activation relate to its underlying structural asymmetries. Particularly, we chose to manually delineate bilateral PTs on the virtually reconstructed cortical surface for each individual, which is very labor intensive but can localize such an anatomically highly variable structure with minimized errors. Moreover, a variety of

macrostructural and microstructural measures were applied to quantify PT structural asymmetries, including surface area, cortical thickness, myelin content, neurite density index (NDI), and orientation dispersion index (ODI). Using these data, we evaluated 1) the functional and structural PT asymmetries at the group level, 2) the functional-structural coupling of PT asymmetries at the individual level, and 3) the within-hemispheric functional-structural coupling of PT at the individual level and its asymmetry while considering the influence of the HG gyrification pattern.

Results

Two experienced raters manually delineated PT and determined the HG gyrification pattern for 907 healthy right-handed young adults (**Fig. 1** [↗](#)).

As shown in [Table 1](#) [↗](#), the manual operation showed excellent interrater reliability and imaging test-retest reproducibility for all PT measures. Among these young adults, left and right HG duplications were identified in 229 and 263 subjects, respectively (occurrence rate: left, 25.3%, right, 29.0%). In terms of the HG gyrification pattern in both hemispheres, all subjects were divided into 4 groups: 503 individuals with bilateral sHGs (L1/R1, 55.5%), 175 individuals with left sHG but right dHG (L1/R2, 19.3%), 141 individuals with left dHG but right sHG (L2/R1, 15.6%), and 88 individuals with bilateral dHGs (L2/R2, 9.7%). Age and sex did not differ among these 4 groups ([Table 2](#) [↗](#)).

For each subject, fMRI activation of the left and right PT during an auditory-language comprehension task was estimated. Both left and right PTs showed significant group-level activation for all three contrasts (i.e., “story – baseline”, “math – baseline”, and “story – math”) within the task (all $T > 15.9$, $P_{FWE} < 0.001$), indicating a strong functional involvement of both PTs in such auditory-language processing. Factor analysis of the activation T values of the three contrasts revealed two factors in each hemisphere: one representing PT functional activation of speech perception and the other representing PT functional activation of speech comprehension.

All functional and structural measures of the left and right PT did not show significant correlation with the score of the two language-related tests in the HCP, i.e., picture vocabulary comprehension and oral reading recognition (all $P_{FWE} > 0.05$).

PT functional and structural asymmetries at the group level

For each of the 4 groups above, PT functional asymmetries of speech perception and comprehension activation were evaluated. As shown in **Fig. 2** [↗](#), significant leftward PT asymmetries were observed for both speech perception and comprehension in the L1/R1 and L1/R2 groups but not in the L2/R1 and L2/R2 groups, suggesting a confounding influence of the HG gyrification pattern on speech-related functional asymmetries of the PT. For each hemisphere, we further compared these PT activations between subjects with sHG and dHG. The results showed that HG duplication was largely accompanied by decreased functional activation in the ipsilateral PT, and the degree of such a decrease varied between the left and right PTs (**Fig. S1** [↗](#)).

Similar analyses were applied to PT structural measures, i.e., surface area, thickness, myelin content, NDI, and ODI. As shown in **Fig. 3** [↗](#), only ODI showed consistent leftward asymmetry across all 4 groups, suggesting a minimal influence of the HG gyrification pattern on the group-level PT asymmetry of this particular measure. In contrast, there was more or less confounding influence of the HG gyrification pattern on the PT asymmetry of the other 4 structural measures. Specifically, we observed 1) significant leftward asymmetry of PT surface area in the L1/R1, L1/R2, and L2/R2 groups but no asymmetry in the L2/R1 group; 2) significant rightward asymmetry of PT thickness in the L1/R1, L2/R1, and L2/R2 groups but no asymmetry in the L1/R2 group; 3)

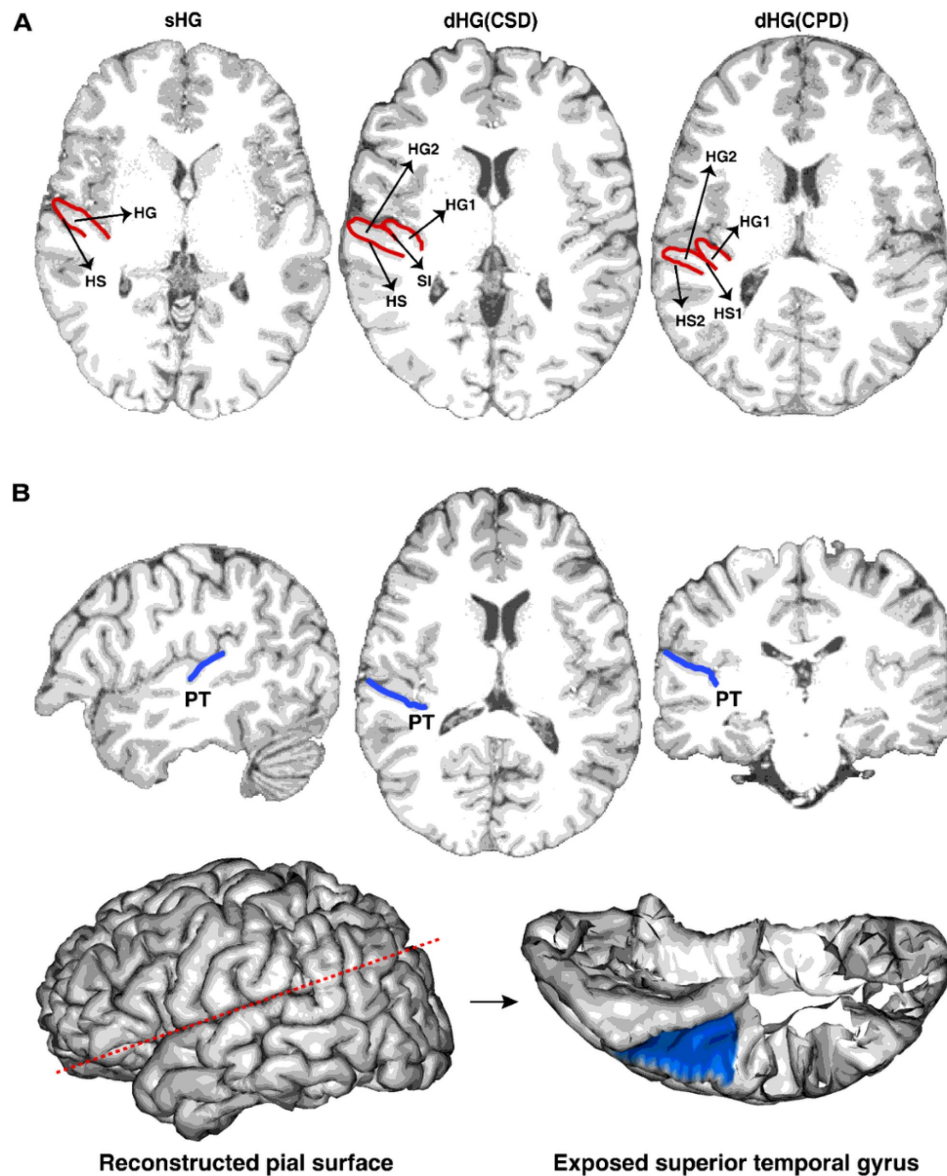


Figure 1

The gyrification pattern of Heschl's gyrus (HG) and manual delineation of the planum temporale (PT) on individual images.

(A) Examples of three types of HG gyrification patterns (red line): single HG (sHG), common stem duplication (CSD), and complete duplication (CPD). HG1, the anterior part of duplicated HG; HG2, the posterior part of duplicated HG; SI, the sulcus intermedius; HS1, the anterior Heschl's sulcus; HS2, the posterior Heschl's sulcus. (B) An example of manually delineated PT on a T1-weighted image (blue line) and reconstructed 32k~ pial surface (blue area). The oblique section plane (red dotted line) was applied to expose the internal surface of the superior temporal gyrus using the Anatomist ⁴².

Table 1

The reliability of PT measures from manually delineated PT

Reliability		Interrater (ICC)	Imaging test-retest (ICC)
PT functional activation	Speech perception	0.99	0.66
	Speech comprehension	0.98	0.81
PT structural measures	Surface area	0.85	0.69
	Thickness	0.97	0.72
	Myelin content	0.98	0.61
	NDI	0.99	0.73
	ODI	0.96	0.86

Table 2

The distribution of HG gyrification patterns

Type of HG gyrification pattern					
	L1/R1	L1/R2	L2/R1	L2/R2	Group Difference
Subject number	503	175	141	88	
Age (mean $\pm$ SD)	28.8 $\pm$ 3.7	28.8 $\pm$ 3.8	29.2 $\pm$ 3.6	28.1 $\pm$ 3.6	$p = 0.19$
Sex (F/M)	275/228	96/79	83/58	55/33	$p = 0.49$

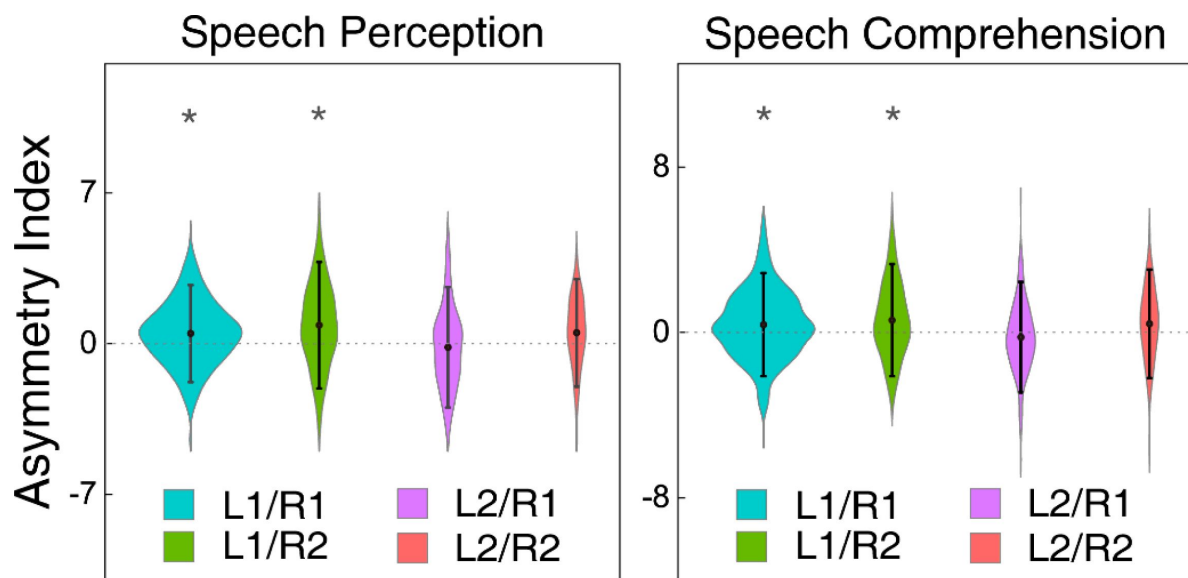


Figure 2

PT functional asymmetry of speech processing at the group level.

L1/R1, single HG on the left and single HG on the right; L1/R2, single HG on the left and duplicated HG on the right; L2/R1, duplicated HG on the left and single HG on the right; L2/R2, duplicated HG on the left and duplicated HG on the right. * denotes Significant difference between left and right PTs (i.e., $P_{FWE} < 0.05$). Positive and negative values of the asymmetry index represent leftward and rightward asymmetry, respectively.

significant leftward and rightward asymmetry of PT myelin content only in the L1/R2 and L2/R1 groups; and 4) significant leftward NDI asymmetry in the L1/R2 group but rightward NDI asymmetry in both the L1/R1 and L2/R1 groups. In addition, the comparison of these PT structural measures between subjects with sHG and dHG indicated that HG duplication was accompanied by a decrease in surface area, myelin content, NDI, and ODI of the ipsilateral PT, with the left and right PT also showing variable degrees of such a decrease (**Fig. S2** [↗](#)).

The asymmetries of functional and structural PT measures were not correlated with the scores of the two language-related tests in the HCP (all $P_{FWE} > 0.05$), as well.

Structure-function associations of PT asymmetries at the individual level

For each functional or structural PT measure, an asymmetry index (AI) was calculated for each subject. For each functional-structural pair of PT measures (2*5 pairs in total), we evaluated the interindividual correlation of their AIs with consideration of the HG gyrification pattern (including a group factor, i.e., L1/R1, L1/R2, L2/R1, or L2/R2). As shown in **Fig. 4** [↗](#), a significant group effect for the correlation between the AIs of PT speech comprehension activation and myelin content was observed, as indicated by a significant interaction with the HG gyrification pattern in the general linear model ($F = 8.83$, $P_{FWE} = 0.03$). According to subsequent post hoc analyses, there was a significant positive correlation between these two AIs in the L1/R1 ($R = 0.12$, $P = 0.01$), L2/R1 ($R = 0.33$, $P = 1.31 \times 10^{-4}$), and L2/R2 ($R = 0.35$, $P = 1.19 \times 10^{-3}$) groups but not in the L1/R2 group. No such significant group effect (i.e., the interaction with group) was observed for any other functional-structural AI pairs.

Among these functional-structural pairs, the functional AI of speech perception activation showed significant positive correlations with the AIs of myelin content ($R = 0.35$, $P_{FWE} = 6.21 \times 10^{-26}$), NDI ($R = 0.16$, $P_{FWE} = 7.59 \times 10^{-5}$), and ODI ($R = 0.27$, $P_{FWE} = 7.44 \times 10^{-14}$), regardless of the HG gyrification pattern. In contrast, functional AI of speech comprehension activation significantly correlated with only the AI of surface area ($R = 0.15$, $P_{FWE} = 8.02 \times 10^{-5}$), regardless of the HG gyrification pattern. Notably, the observed interindividual correlations were consistently positive, strongly supporting an individual-level coupling of PT functional and structural asymmetries.

Within-hemispheric PT structure-function associations at the individual level

In addition to PT functional and structural AIs, we evaluated the correlations between PT functional and structural measures of each hemisphere per se separately (**Fig. 5** [↗](#)). For speech perception, functional activation correlated positively with myelin content and ODI for either the left (myelin content: $R = 0.16$, $P_{FWE} = 5.63 \times 10^{-5}$; ODI: $R = 0.16$, $P_{FWE} = 8.21 \times 10^{-5}$) or right PT (myelin content: $R = 0.24$, $P_{FWE} = 3.68 \times 10^{-11}$; ODI: $R = 0.17$, $P_{FWE} = 3.59 \times 10^{-5}$), regardless of the HG gyrification pattern. The Fisher Z test, however, showed no significant difference in the degree of functional-structural correlations between the left and right PTs (myelin content: $Z = -1.67$, $P = 0.10$; ODI: $Z = -0.04$, $P = 0.97$). Moreover, speech perception activation showed a significant positive correlation with NDI for the right PT ($R = 0.16$, $P_{FWE} = 1.38 \times 10^{-4}$) but not the left PT.

Regarding speech comprehension, there was a significant effect of the HG gyrification pattern on its correlation with myelin content for the left PT, i.e., a significant interaction term in the general linear model ($F = 10.8$, $P_{FWE} = 0.02$). Subsequent post hoc analyses revealed a significant correlation across subjects with a dHG ($R = 0.27$, $P = 5.22 \times 10^{-5}$) but not with a sHG ($R = 0.04$, $P = 0.27$) in the left hemisphere. In contrast, speech comprehension activation positively correlated with myelin content for the right PT ($R = 0.21$, $P_{FWE} = 1.28 \times 10^{-8}$), regardless of the HG gyrification pattern. Finally, we also observed a significant correlation between functional activation of speech comprehension and NDI for the right PT ($R = 0.19$, $P_{FWE} = 2.33 \times 10^{-6}$) but not for the left PT.

Figure 3

PT structural asymmetry at the group level.

L1/R1, single HG on the left and single HG on the right; L1/R2, single HG on the left and duplicated HG on the right; L2/R1, duplicated HG on the left and single HG on the right; L2/R2, duplicated HG on the left and duplicated HG on the right. NDI, neurite density index; ODI, orientation dispersion index. * denotes significant difference between left and right PTs ($P_{FWE} < 0.05$). Positive and negative values of the asymmetry index represent leftward and rightward asymmetry, respectively.

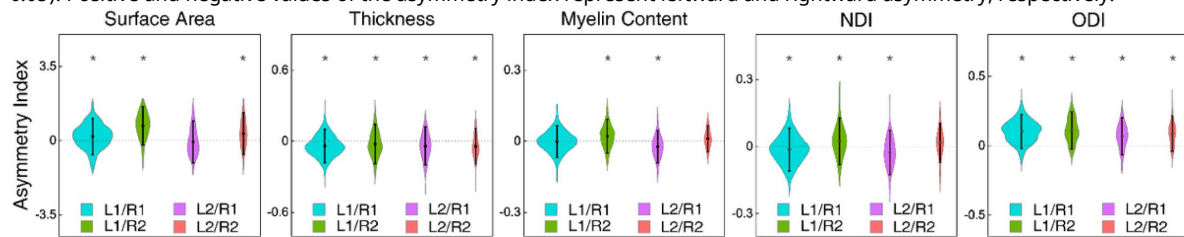
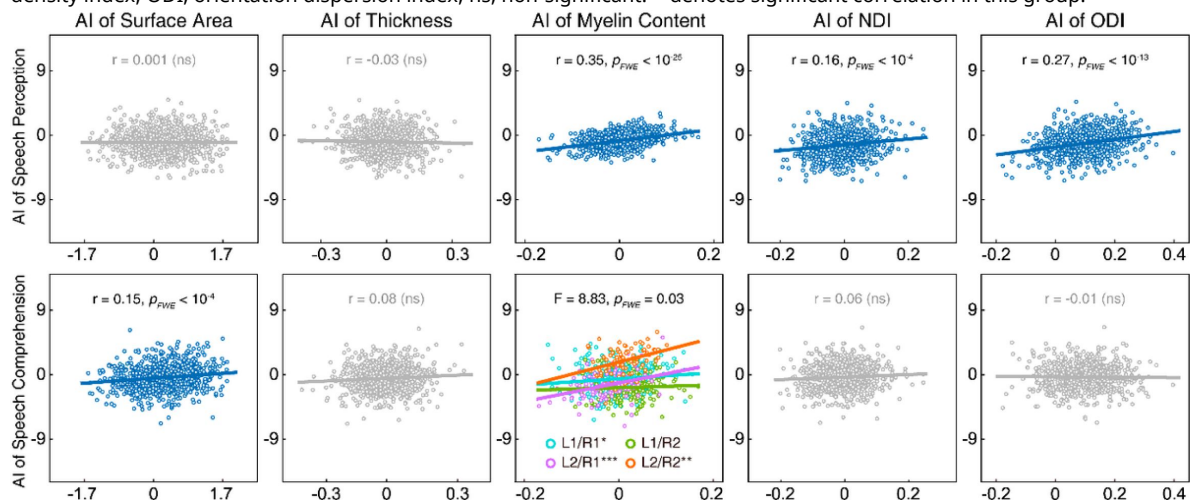


Figure 4

Interindividual correlation of PT functional and structural asymmetries.

The correlation of the AI of speech comprehension activation and the AI of myelin content showed significant dependence on the gyrification pattern of HG (i.e., a significant interaction with the group factor); therefore, scatter plots for their correlations within the 4 groups were displayed separately. L1/R2, single HG on the left and duplicated HG on the right; L2/R1, duplicated HG on the left and single HG on the right; L2/R2, duplicated HG on the left and duplicated HG on the right. For the other pairs of functional and structural AIs, the gyrification pattern of HG showed no significant effect on their correlations. The scatter plots for nonsignificant correlations ($P_{FWE} > 0.05$) are colored gray. AI, asymmetry index; NDI, neurite density index; ODI, orientation dispersion index; ns, non-significant. * denotes significant correlation in this group.



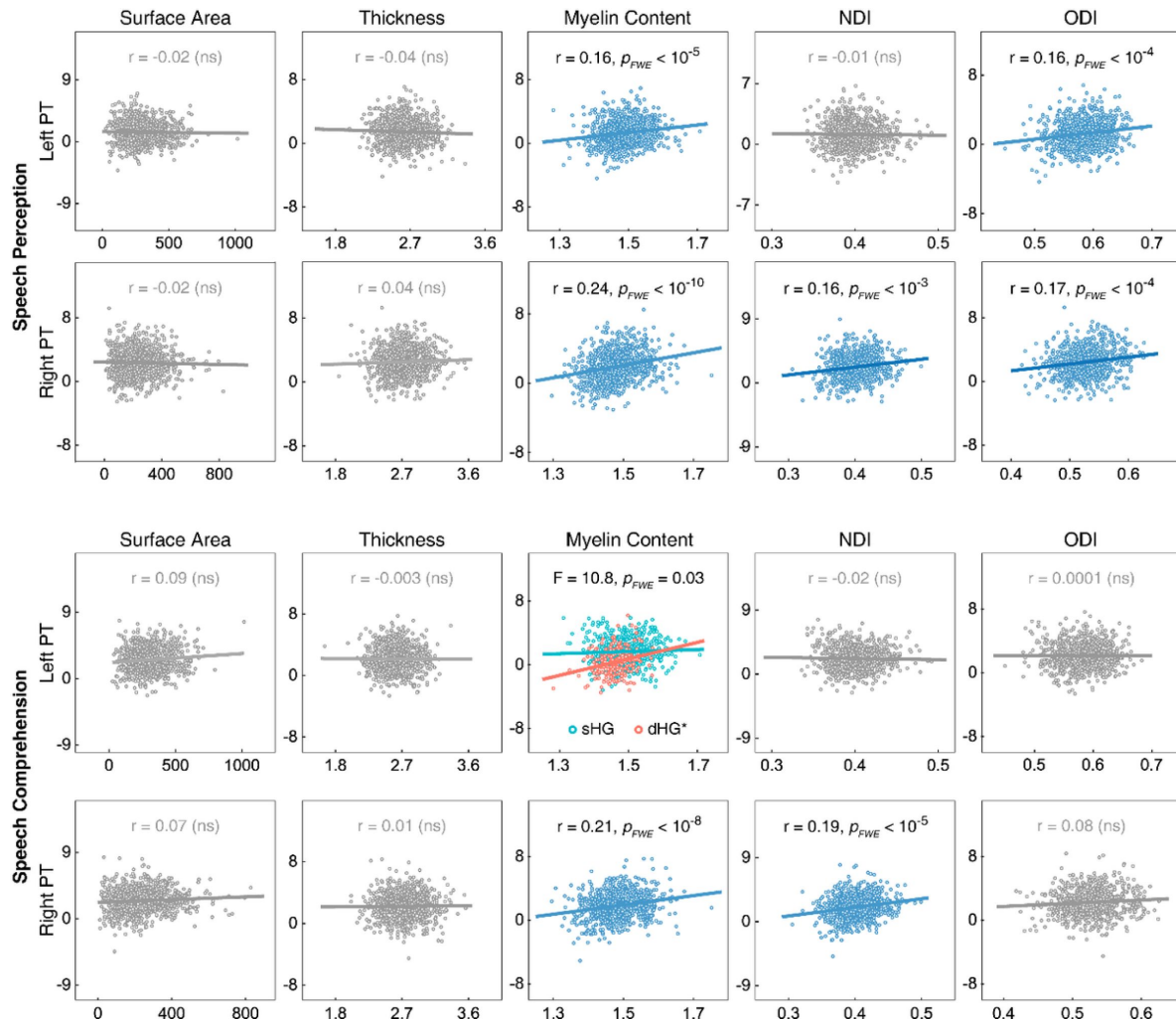


Figure 5

Interindividual functional-structural correlations of the left and right PT.

The correlation of speech comprehension activation and myelin content for the left PT showed significant dependence on the gyrification pattern of HG (i.e., a significant interaction with the group factor); therefore, scatter plots for their correlations within the 2 groups were displayed separately. sHG, single Heschl's gyrus; dHG, duplicated Heschl's gyrus. For the other pairs of functional and structural measures of the left and right PT, the gyrification pattern of HG showed no significant effect on their correlations. The scatter plots for nonsignificant correlations ($P_{FWE} > 0.05$) are colored gray. NDI, neurite density index; ODI, orientation dispersion index; ns, non-significant. * denotes significant correlation in this group.

Specificity of the observed PT structure-function associations

For the included subjects, another 6 task-based fMRI scans were acquired. To test whether the observed PT functional-structural couplings above are specific to auditory-language processing, we further estimated left and right PT activation of the main contrast from these 6 tasks. As shown in [Table S1](#), only 3 tasks, i.e., incentive processing, social cognition, and relational processing, showed significant group-level activation in both PTs (though much weaker than the language processing above: $3.18 < T < 14.5$, $10^{-12} < P_{FWE} < 0.014$), suggesting the functional involvement of PT in these tasks. We therefore repeated all coupling analyses above in these 3 tasks. For any of the 3 tasks, no significant interindividual correlation was observed between functional and structural AIs or between functional and structural measures within each hemisphere ([Table S3](#)–[S6](#)). These negative results strongly indicate the specificity of the observed PT functional-structural couplings to the functional activation of auditory-language processing.

In addition, to evaluate the spatial specificity of our observed PT functional-structural couplings, we estimated the functional activation of speech perception and comprehension of the entire hemisphere from the main auditory-language processing task. The coupling analysis showed no significant interindividual correlation between PT structural AIs and speech-related functional AIs of the entire hemisphere or between PT structural measures and speech-related functional activation of the entire ipsilateral hemisphere ([Table S7](#)–[S10](#)). Therefore, PT structural measures or their asymmetries were coupled with PT-specific functional activation or their asymmetries rather than with general functional activation or its asymmetry in auditory-language processing.

Discussion

Using a large cohort of healthy adults with high-quality multimodal MRI data and highly accurate PT determination, the present study shows clear associations between structural and functional asymmetries in the PT. Crucially, most effects were observed for asymmetries in microstructural features of the PT, such as myelin content, neurite density, or neurite orientation dispersion. This highlights that microstructural asymmetries are likely more relevant for functional asymmetries than macrostructural asymmetries. For the left or right PT per se, its functional activation of speech perception and comprehension also correlated with its myelin content, neurite density, and neurite orientation dispersion, both selectively and hemisphere-dependently. Among the observed PT functional-structural associations, only those between speech comprehension activation and myelin content showed interaction with the HG gyrification pattern. Finally, these observed functional-structural associations are highly specific to the within-PT functional activation of auditory-language processing.

As one of the most prominent structural asymmetries across the entire human brain, PT structural asymmetries have been well recognized and widely believed to play a critical role in human auditory or language processing [8](#), [19](#), [21](#), [30](#), [43](#)–[45](#). In concordance, anomalies of PT asymmetries in surface area or thickness have been reported in various brain disorders with auditory or language deficits, e.g., dyslexia, autism, and schizophrenia [17](#), [21](#), [46](#)–[52](#). To understand the role of PT structural asymmetries in language, it is critical to determine how they relate to functional patterns of language-related processing, e.g., language-related functional asymmetries. Some evidence has demonstrated an association of PT structural asymmetry with language-related functional asymmetries, but these functional asymmetries were not measured directly from the PT [16](#), [27](#), [28](#), [53](#). For instance, PT asymmetry in surface area was correlated with regional asymmetries of language-related functional activation around the Sylvian fissure but not the activation of the PT [16](#).

Our present study provides the first empirical evidence of interindividual associations between functional and structural asymmetries within the PT, implying a functional pathway from PT structural asymmetry to PT functional asymmetry to effective speech processing. A large sample size and accurate PT localization based on well-trained manual operation were crucial in obtaining these results. For other PT-related studies with large sample sizes, however, it is difficult to apply such a labor-intensive manual PT labeling approach, and therefore, an accurate automatic labeling approach should be developed. Notably, our observed PT structural-functional associations were across healthy right-handed young adults, and it is unclear whether these associations could be extrapolated to children, left-handed adults, or patients. Therefore, future studies are required to evaluate the functional-structural association of PT asymmetries in other populations, which should provide insight into the modulating factor of functional-structural associations of PT.

In the context of brain asymmetries, our currently observed functional-structural coupling of PT asymmetries at the individual level empirically proves the role of structural gray matter asymmetries in the functional lateralization of the same brain area, supporting the compelling hypothesis that “functional lateralization is guided by structural asymmetry”³. For a specific brain area, however, its gray matter asymmetries are unlikely to be the only determinant for its functional lateralization. The relevant corpus callosum (interhemispheric connection) and structural white matter asymmetries should also play important roles in its functional lateralization, and integrating the three factors is encouraged for detangling structural mechanisms underlying functional lateralization^{2,54–57}.

The observed functional-structural coupling of PT asymmetries strongly depends on functional and structural measures. On the functional side, only PT functional activation of auditory speech processing showed such coupling with PT structural asymmetries, indicating the functional and spatial specificity of PT asymmetry in auditory-language processing^{8,13,58,59}. Speech perception and comprehension also showed functional-structural coupling with different structural measures, compatible with the dissociation of these two speech-processing components^{15,60}. It is possible that hemispheric asymmetries of different functional processing are generally related to distinct structural mechanisms. On the structural side, the observed functional-structural coupling of PT asymmetries mainly involves microstructural measures, e.g., myelin content, neurite density, and neurite orientation dispersion, except for the association of speech comprehension with surface area. These microstructural measures represent specific aspects of cortical composition on the level of axons or dendrites^{61,62}, therefore implying a dominant role of within-PT microconnectional or microcircuitry asymmetries in functional asymmetries of PT. Specific microstructural asymmetry between bilateral PTs might cause less recruitment of the nondominant hemisphere in particular speech processing, therefore resulting in more pronounced functional lateralization^{16,44,53}.

Compatibly with the functional-structural coupling of PT asymmetries, only microstructural measures, including myelin content, neurite density, and neurite orientation dispersion, showed an association with speech-related activation for the left or right PT *per se*. Similar functional-structural associations have been observed in other non-PT cortical areas, e.g., between myelin content and task-evoked activities^{63–68}. These findings together indicate an important contribution of intracortical microcircuits to functional activity: cortical areas with greater intracortical fiber density and more complex dendritic structures are likely accompanied by stronger local activation during the task. Intriguingly, within-hemispheric PT functional-structural coupling could be asymmetric, e.g., existing only in one hemisphere but not in the other. For instance, EEG-measured neurophysiological processing of speech perception correlated with neurite density of the left PT but not the right PT, as recently revealed by Ocklenburg and colleagues³⁰. Consistent with this, within-hemispheric PT functional-structural coupling of neurite density was also asymmetric in our study: only in the right PT but not in the left PT. These

particular asymmetries highlight the distinct role of PT neurite density in speech processing between the two hemispheres, but computational implementation for such hemisphere-dependent functional-structural association warrants further investigation.

The functional-structural coupling of PT asymmetries should relate to the asymmetry of within-hemispheric functional-structural coupling between the left or right PT *per se* but in a complex manner. The coupling of PT asymmetries could be accompanied by either asymmetric (e.g., speech perception vs. neurite density) or nonasymmetric within-hemispheric functional-structural couplings (e.g., speech perception vs. myelin content or neurite orientation dispersion). On the other hand, asymmetric within-hemispheric structure-function association does not necessarily lead to a coupling of PT asymmetries (e.g., speech comprehension activation vs. neurite density). Finally, PT functional and structural asymmetries could be associated, even though there was no within-hemispheric functional-structural coupling for both left and right PTs (e.g., surface area vs. speech comprehension activation). This implied that PT asymmetries and their functional-structural coupling might represent unique structural and functional information, independent of within-hemispheric PT measures *per se*. Overall, there was no simple causal relationship between these functional-structural couplings, and they likely capture distinct aspects of PT functional-structural associations and therefore should be investigated separately.

In line with previous observations, HG duplication occurred considerably in healthy adults in the present study and induced significant changes in both PT functional and structural measures as well as their asymmetries ^{33,41}. However, the vast majority of functional-structural coupling of both PT asymmetries and within-hemispheric PT measures were not significantly affected by the HG gyrification pattern. Therefore, HG duplication-induced individual changes in PT measures or their asymmetry do not necessarily lead to changes in the interindividual association between these measures or asymmetries, suggesting the robustness of these PT functional-structural couplings. Notably, the left HG duplication did show a significant impact on the coupling of speech comprehension activation and myelin content of the left PT as well as on their asymmetry coupling. This could be related to functional reorganization of the left PT when a left duplication of HG occurs, but this speculation should be empirically tested in future studies. Given these observed influences of the duplication of HG as well as its substantial incidence in healthy adults, more attention should be given to this phenomenon, particularly when studying HG or PT and its asymmetry.

In conclusion, the association between specific PT functional and microstructural asymmetries provides direct empirical support for the contribution of structural asymmetry to functional lateralization of the same cortical area. Moreover, the findings highlight a critical role of microstructural PT asymmetries in auditory-language processing.

Materials And Methods

Participants

In the present study, all participants of the human connectome project (HCP young adult, S1200 release) were included. The project was reviewed and approved by the Institutional Ethics Committee of Washington University in St. Louis, Missouri. The HCP young adult cohort consists of healthy individuals without neurodevelopmental, neuropsychiatric, or neurologic disorders. All participants signed written informed consent forms. For more details, please refer to Van Essen et al. ⁶⁹.

Due to quality issues (quality control codes A and B from the HCP minimal preprocessing pipeline), 72 subjects were excluded. To control for the potential confounding effect of handedness, we included only qualified right-handed subjects (907 in total, Edinburgh Handedness questionnaire >

20) in the analysis.

Behavioral tests were applied to HCP individuals, including two language-related tests: picture vocabulary comprehension and oral reading recognition. The age-adjusted scores for these two language-related tests were used in the current study.

MRI acquisition and preprocessing

MRI data of all subjects were collected using the same 3T Siemens Skyra magnetic resonance machine at Washington University in St. Louis with a 32-channel head coil⁶⁹. Briefly, T1-weighted images were acquired by using a magnetized rapid gradient-echo imaging (MPRAGE) sequence with the following parameters: repetition time (TR) = 2400 ms, echo time (TE) = 2.14 ms, reversal time (TI) = 1000 ms, flip angle (FA) = 8°, field of view (FOV) = 224 × 224 mm², voxel size 0.7 mm isotropic. T2-weighted images were acquired using the variable flip angle turbo spin-echo (SPACE) sequence with the following parameters: TR = 3200 ms, TE = 565 ms, field of view (FOV) = 224 × 224 mm², voxel size 0.7 mm isotropic.

Diffusion-weighted images were acquired with spin-echo EPI with the following parameters: TR = 5520 ms, TE = 89.5 ms, field of view (FOV) = 210 × 180 mm², voxel size 1.25 mm isotropic, 111 slices, 90 directions for each of three shells of b-values (b = 1000, 2000 and 3000 s/mm²) and 18 nondiffusion-weighted (b = 0 s/mm²) volumes.

Task-fMRI scans were acquired under 7 tasks: language processing, working memory, incentive processing, relational processing, motor, social cognition, and emotional processing. The language processing task consisted of two runs containing 4 blocks of an auditory story task and 4 length-match blocks of an auditory math task. For detailed paradigms of all these tasks, please refer to Barch et al., 2013⁷⁰. All fMRI data were acquired by using a gradient-echo echo planar imaging (GE-EPI) sequence with the following parameters: TR = 720 ms, TE = 33.1 ms, FA = 2°, FOV = 208 × 180 mm², voxel size 2 mm isotropic, 72 slices, multiband accelerated factor = 8.

All MRI images were preprocessed using the HCP minimal preprocessing pipeline⁷¹. For each subject, the HCP minimal preprocessing pipeline provides the native pial and white surfaces that are resampled onto the standard 32k_fs_LR mesh (~ 32k vertices for each hemispheric surface).

Manually delineating PT and determining the HG gyrification pattern

We followed a well-established procedure for manually delineating PT and determining HG gyrification patterns^{46,52}. The procedure was carried out using the Anatomist software platform embedded in BrainVISA, a sophisticated visualization and labeling tool⁷². The Anatomist software platform allows for simultaneously localizing a given coordinate on the surface as well as on the coronal, axial, and sagittal views of the T1 image.

Two well-trained raters (Q.P. and Z.G.) blinded to the subjects' demographics carefully examined the native T1 image for each subject and determined the HG gyrification pattern according to the widely used criterion⁷³. As illustrated in **Fig. 1**, there were three types of HG gyrification patterns. The first is the single HG (sHG), i.e., only one transverse gyrus on the superior temporal gyrus (STG). The second is common stem duplication (CSD): the sulcus intermedius of Beck (SI) splits the HG by at least half of the length but never extends to the internal border of the gyrus. The last is complete posterior duplication (CPD): the HG is medially split into two separate parts by an additional Heschl's sulcus (HS1). In the present study, both CSD and CPD were classified as duplicated HG (dHG).

Next, the two raters performed the PT delineation on the native pial surface for each subject while simultaneously viewing the coronal, axial, and sagittal slices of the T1 image. According to Altarelli et al., 2014⁴⁶, the anterior border of the PT was indicated by Heschl's sulcus or the second Heschl's sulcus; the posterior border of the PT was indicated by a change in the slope of the continuous plane characterizing the planum on the coronal view; and the lateral border was defined as the most lateral margin of the STG. For more details, please refer to Altarelli et al., 2014 and Vanderauwera et al., 2018^{46,52}.

PT functional activation

The HCP minimal preprocessing pipeline provides individual-level T activation maps on the 32k_fs_LR cortical surface for all contrasts between task conditions. For each subject, we extracted the averaged T value across PT vertices of each hemisphere for each of the three contrasts under the language task: “story - baseline”, “math - baseline”, and “story - math”. Significant correlations across individuals were observed between the T values of these contrasts: “story - baseline” vs. “math - baseline” (Left: $r = 0.94$, $P_{FWE} < 0.001$; Right: $r = 0.95$, $P_{FWE} < 0.001$), “story - baseline” vs. “story - math” (Left: $r = 0.50$, $P_{FWE} < 0.001$; Right: $r = 0.42$, $P_{FWE} < 0.001$), “math - baseline” vs. “story - math” (Left: $r = 0.19$, $P_{FWE} < 0.001$; Right: $r = 0.13$, $P_{FWE} = 0.001$). We then performed exploratory factor analysis separately for each hemisphere, and obtained two factors: one representing functional activation of speech perception (loadings in left hemisphere: “story - baseline”, 0.94; “math - baseline”, 1.00; “story - math”, 0.16; loadings in right hemisphere: “story - baseline”, 0.96; “math - baseline”, 1.00; “story - math”, 0.15) and the other representing functional activation of speech comprehension (loadings in left hemisphere: “story - baseline”, 0.39; “math - baseline”, 0.06; “story - math”, 0.99; loadings in right hemisphere: “story - baseline”, 0.32; “math - baseline”, 0.01; “story - math”, 0.99). The two factor scores were then used in subsequent statistical analyses.

In the first specificity analysis, we similarly extracted the averaged T value across PT vertices of each hemisphere for the main contrast under the other 6 tasks.

In the second specificity analysis, we extracted the averaged T value across the entire hemisphere for each of the three contrasts under the language task. As above, two factor scores representing functional activation of speech perception and comprehension of the entire hemisphere were estimated and applied in statistical analyses.

PT structural measures

For each subject, we directly obtained cortical maps of surface area, thickness, and T1w/T2w ratio-based myelin content from the HCP minimal preprocessing pipeline. For each delineated PT, the total surface area and averaged thickness and myelin content across PT vertices were calculated. For each subject, we also calculated the neurite density index (NDI) and orientation dispersion index (ODI) using a diffusion MRI-based neurite orientation dispersion and density imaging (NODDI) approach⁶². Specifically, voxelwise values for these two measures were first estimated using AMICO⁷⁴ and then projected onto the PT vertices using the ribbon mapping method⁷⁵. For each delineated PT, we averaged the values of the two measures across PT vertices.

Interrater reliability and test-retest reproducibility

To assess interrater reliability, the two raters both determined the HG gyrification pattern and delineated the PT for the same 20 randomly selected subjects. The results of the HG gyrification pattern reached 100% consistency. Regarding the PT measures, we calculated the intraclass correlation coefficient (ICC). As shown in Table 2⁴⁷, the interrater ICC values ranged from 0.85 to 0.99, indicating excellent interrater reliability for these measures.

To evaluate the test-retest reproducibility of both manual operation and brain imaging, one rater (P.Q.) further determined the HG gyrification pattern and delineated the PT for the 43 test-retest HCP subjects who were rescanned (test-retest interval: 1–11 months). The test-retest results of the HG gyrification pattern also reached 100% consistency. As shown in [Table 1](#), the imaging test-retest ICC values ranged from 0.61 to 0.86, indicating excellent test-retest reproducibility.

Asymmetry index (AI)

To quantify PT asymmetries, an asymmetry index (AI) was calculated for each subject as follows: functional AI = (Left – Right); structural AI = (Left – Right)/[(Left + Right)/2]. We excluded the denominator [(Left + Right)/2] in the functional AI because the values of functional activation could be negative and therefore could result in an inappropriate scaling of the numerator (Left – Right) rather than the originally expected normalization to [-1, 1].

Statistical analysis

In terms of the HG gyrification pattern in both hemispheres, all subjects were divided into 4 groups: bilateral sHGs (L1/R1), left sHG but right dHG (L1/R2), left dHG but right sHG (L2/R1), and bilateral dHGs (L2/R2). For each group, we tested the hemispheric asymmetry of each PT functional activation and structural measure. A linear mixed model (LME) was used, in which “hemisphere” was a fixed effect and “individual identity” was a random effect. In the model, age, sex, and hemispheric brain volume⁷⁶ were included as covariates. To further evaluate the influence of the HG gyrification pattern on functional and structural measures of the left and right PT per se, we divided all subjects into two groups for each hemisphere: sHG or dHG. The PT measures of the two groups were then compared using two-sample t tests after controlling for age, sex, and hemispheric brain volume.

We then evaluated whether PT functional activation, structural measures (2*14 in total), and their AIs (2*7 pairs in total) correlated with the scores of picture vocabulary comprehension and oral reading recognition tests, wherein age, sex, HG gyrification type and hemispheric or total brain volume (ICV) were covariates.

Next, to determine whether PT structural asymmetries relate to functional asymmetries at the individual level, we applied a general linear model (GLM) to each pair of functional and structural AIs (2*5 pairs in total). Specifically, the model includes “functional AI” as the response variable and “structural AI”, “group” (i.e., L1/R1, L1/R2, L2/R1, or L2/R2), and “structural AI × group” as predictor variables, wherein age, sex, and intracranial volume (ICV) were covariates. As described previously^{77–79}, we first evaluated whether there was a significant “structural AI × group” interaction, i.e., whether the correlation between functional and structural AIs differed between the 4 groups. If not, the term “structural AI” was assessed after excluding the interaction term; if yes, post hoc GLM analysis was conducted to assess the term “structural AI” for each of the 4 groups.

Last, we evaluated the correlation of PT functional activation with its structural measures within each hemisphere (4*5 pairs in total) and its asymmetry between the two hemispheres. Specifically, for each hemisphere, we applied a GLM with “functional activation” as the response variable and “structural measure”, “group” (i.e., sHG or dHG), and “structural measure × group” as predictor variables, wherein age, sex, and hemispheric brain volume (ICV) were covariates. Again, we first evaluated whether there was a significant “structural measure × group” interaction. If not, the term “structural measure” was assessed after excluding the interaction term; if yes, a post hoc GLM analysis was conducted to assess the term “structural measure” for each group. For each pair of functional activation and structure measures showing a significant correlation for both the left and right PT, we then applied Fisher-Z tests to determine whether the functional-structural correlations differed between bilateral PTs.

For each type of statistical analysis above, the significance level was set as $p < 0.05$ after Bonferroni correction.

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Declarations

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Investigation: PQ, QB, ZG, LY, HL, PL, XL, JL, XK, YX

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Writing—review & editing: GG, PQ, QB, SO, ZG, LY, HL, PL, XL, JL, XK, YX, BS

Data and materials availability

The datasets analysed during the current study are available at <https://www.humanconnectome.org/> . In addition, processed data and code may be requested from the authors.

Supplementary Files

Fig. S1

The difference in PT functional activation between subjects with single and duplicated HG.

sHG, single Heschl's gyrus; dHG, duplicated Heschl's gyrus; ns, non-significant. Nonsignificant differences ($p_{FWE} > 0.05$) are colored gray.

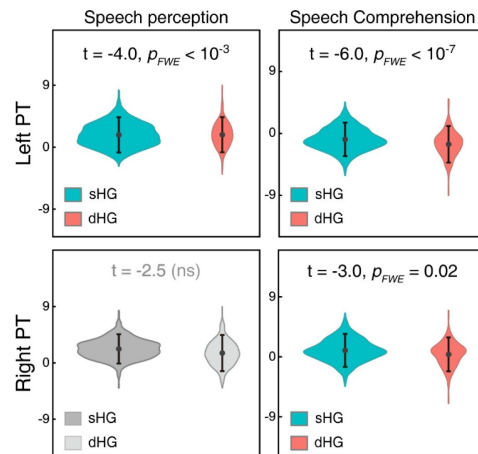


Fig. S2

The difference in PT structural measures between subjects with single and duplicated HG.

sHG, single Heschl's gyrus; dHG, duplicated Heschl's gyrus. Nonsignificant differences ($p_{FWE} > 0.05$) are colored gray. NDI, neurite density index; ODI, orientation dispersion index; ns, non-significant.

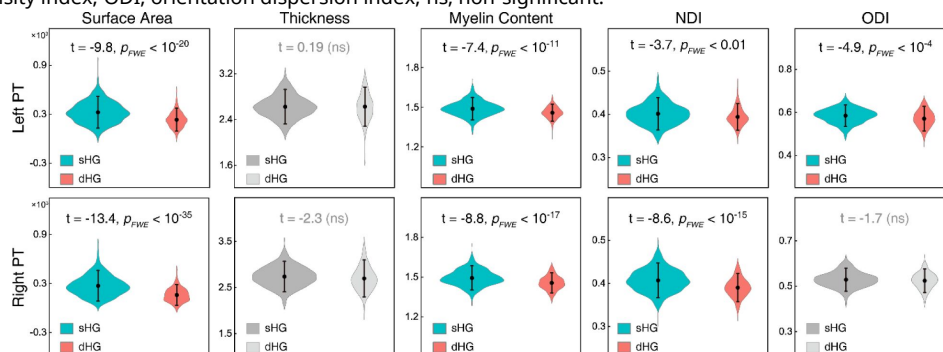


Table S1

Group-level activation of main contrasts for the HCP fMRI tasks.

fMRI task	Contrast	Left PT		Right PT	
		<i>T</i>	<i>P_{FWE}</i>	<i>T</i>	<i>P_{FWE}</i>
Language processing	“math – baseline”	43.6	< 0.001 [*]	31.5	< 0.001 [*]
	“story – baseline”	45.9	< 0.001 [*]	33.6	< 0.001 [*]
	“story – math”	22.0	< 0.001 [*]	15.9	< 0.001 [*]
Working memory	“2Back – 0Back”	-10.8	> 0.99	-13.3	> 0.99
Incentive processing	“punish – reward”	3.18	0.014 [*]	3.37	0.007 [*]
Emotion processing	“faces – shapes”	-5.12	> 0.99	-9.56	> 0.99
Social cognition	“random – tom”	9.47	< 0.001 [*]	11.0	< 0.001 [*]
Motor	“AVG – baseline”	5.59	< 0.001 [*]	0.05	> 0.99
Relational processing	“match – rel”	14.5	< 0.001 [*]	14.0	< 0.001 [*]

Table S2

Group-level hemispheric asymmetry of PT functional and structural measures

	L1/R1		L1/R2		L2/R1		L2/R2	
	<i>T</i>	<i>P_{FWE}</i>	<i>T</i>	<i>P_{FWE}</i>	<i>T</i>	<i>P_{FWE}</i>	<i>T</i>	<i>P_{FWE}</i>
Speech perception	-7.05	< 0.001 [*]	-5.75	< 0.001 [*]	1.10	> 0.99	-1.98	> 0.99
Speech comprehension	-5.04	< 0.001 [*]	-4.77	< 0.001 [*]	1.83	> 0.99	-1.42	> 0.99
Surface area	-6.83	< 0.001 [*]	-14.4	< 0.001 [*]	2.06	> 0.99	-4.88	< 0.001 [*]
Thickness	10.0	< 0.001 [*]	3.08	0.06	4.39	< 0.001 [*]	3.58	0.01
Myelin content	0.99	> 0.99	-5.97	< 0.001 [*]	6.50	< 0.001 [*]	-2.67	0.23
NDI	4.43	< 0.001 [*]	-4.26	< 0.001 [*]	4.83	< 0.001 [*]	-2.59	0.29
ODI	-26.7	< 0.001 [*]	-15.8	< 0.001 [*]	-8.69	< 0.001 [*]	-9.17	< 0.001 [*]

Table S3

The interaction effect of “PT structural AI × HG gyrification pattern” on PT nonspeech-related functional AIs (the first specificity analysis)

	Social cognition		Incentive processing		Relational processing	
	<i>F</i>	<i>P_{FWE}</i>	<i>F</i>	<i>P_{FWE}</i>	<i>F</i>	<i>P_{FWE}</i>
Surface area	0.64	> 0.99	1.61	> 0.99	0.42	> 0.99
Thickness	1.48	> 0.99	0.06	> 0.99	0.24	> 0.99
Myelin content	0.21	> 0.99	2.75	> 0.99	0.22	> 0.99
NDI	0.40	> 0.99	0.27	> 0.99	0.19	> 0.99
ODI	0.66	> 0.99	2.46	> 0.99	2.57	> 0.99

Table S4

The correlations of PT structural AIs with PT nonspeech-related functional activation AIs after controlling for the HG gyrification pattern (the first specificity analysis)

	Social cognition		Incentive processing		Relational processing	
	<i>R</i>	<i>P_{FWE}</i>	<i>R</i>	<i>P_{FWE}</i>	<i>R</i>	<i>P_{FWE}</i>
Surface area	-0.07	> 0.99	-0.03	> 0.99	0.04	> 0.99
Thickness	-0.02	> 0.99	0.04	> 0.99	-0.01	> 0.99
Myelin content	0.04	> 0.99	0.03	> 0.99	-0.09	0.16
NDI	0.01	> 0.99	-0.01	> 0.99	-0.09	0.25
ODI	0.07	0.95	0.04	> 0.99	0.01	> 0.99

Table S5

The interaction effect of “PT structural measure × HG gyrification pattern” on PT nonspeech-related functional activation for each hemisphere (the first specificity analysis)

		Social cognition		Incentive processing		Relational processing	
		<i>F</i>	<i>P_{FWE}</i>	<i>F</i>	<i>P_{FWE}</i>	<i>F</i>	<i>P_{FWE}</i>
Left	Surface area	2.74	> 0.99	0.67	> 0.99	0.53	> 0.99
	Thickness	0.34	> 0.99	0.98	> 0.99	0.53	> 0.99
	Myelin content	4.61	0.96	2.57	> 0.99	1.24	> 0.99
	NDI	2.39	> 0.99	0.59	> 0.99	0.03	> 0.99
	ODI	0.33	> 0.99	0.14	> 0.99	0.38	> 0.99
Right	Surface area	0.02	> 0.99	0.39	> 0.99	0.39	> 0.99
	Thickness	0.04	> 0.99	2.56	> 0.99	5.00	0.77
	Myelin content	0.11	> 0.99	0.39	> 0.99	0.33	> 0.99
	NDI	2.54	> 0.99	0.27	> 0.99	0.25	> 0.99
	ODI	1.69	> 0.99	2.48	> 0.99	0.05	> 0.99

Table S6

The correlations of PT structural measures with PT nonspeech-related functional activation for each hemisphere after controlling for the HG gyrification pattern (the first specificity analysis)

		Social cognition		Incentive processing		Relational processing	
		<i>R</i>	<i>P_{FWE}</i>	<i>R</i>	<i>P_{FWE}</i>	<i>R</i>	<i>P_{FWE}</i>
Left	Surface area	0.04	> 0.99	-0.06	> 0.99	0.05	> 0.99
	Thickness	0.02	> 0.99	0.01	> 0.99	-0.07	> 0.99
	Myelin content	0.06	> 0.99	0.01	> 0.99	-0.05	> 0.99
	NDI	-0.002	> 0.99	-0.05	> 0.99	-0.04	> 0.99
	ODI	0.03	> 0.99	-0.03	> 0.99	-0.02	> 0.99
Right	Surface area	-0.09	0.33	0.00	> 0.99	-0.02	> 0.99
	Thickness	-0.04	> 0.99	0.01	> 0.99	0.01	> 0.99
	Myelin content	0.01	> 0.99	0.01	> 0.99	-0.09	0.32
	NDI	0.03	> 0.99	0.01	> 0.99	-0.09	0.28
	ODI	-0.002	> 0.99	0.05	> 0.99	-0.01	> 0.99

Table S7

The interaction effect of “PT structural AI × HG gyrification pattern” on speech-related functional AIs of the entire hemisphere (the second specificity analysis)

		Entire hemisphere			
		Speech perception		Speech comprehension	
		<i>F</i>	<i>P_{FWE}</i>	<i>F</i>	<i>P_{FWE}</i>
PT	Surface area	0.28	> 0.99	0.87	> 0.99
	Thickness	0.13	> 0.99	0.02	> 0.99
	Myelin content	1.77	> 0.99	0.06	> 0.99
	NDI	0.03	> 0.99	1.68	> 0.99
	ODI	0.94	> 0.99	0.27	> 0.99

Table S8

The correlations of PT structural AIs with speech-related functional AI of the entire hemisphere after controlling for the HG gyrification pattern (the second specificity analysis)

		Entire hemisphere			
		Speech perception		Speech comprehension	
		<i>R</i>	<i>P_{FWE}</i>	<i>R</i>	<i>P_{FWE}</i>
PT	Surface area	-0.07	0.42	0.03	> 0.99
	Thickness	-0.07	0.50	-0.05	> 0.99
	Myelin content	0.07	0.54	0.04	> 0.99
	NDI	-0.02	> 0.99	0.03	> 0.99
	ODI	0.04	> 0.99	-0.01	> 0.99

Table S9

The interaction effect of “PT structural measure × HG gyrification pattern” on speech-related functional activation of the entire ipsilateral hemisphere (the second specificity analysis)

		Entire ipsilateral hemisphere			
		Speech perception		Speech comprehension	
		<i>F</i>	<i>P_{FWE}</i>	<i>F</i>	<i>P_{FWE}</i>
Left PT	Surface area	7.45	0.13	1.46	> 0.99
	Thickness	0.00	> 0.99	0.04	> 0.99
	Myelin content	0.12	> 0.99	0.18	> 0.99
	NDI	0.33	> 0.99	0.20	> 0.99
	ODI	0.51	> 0.99	2.35	> 0.99
Right PT	Surface area	0.37	> 0.99	0.29	> 0.99
	Thickness	0.68	> 0.99	0.46	> 0.99
	Myelin content	0.18	> 0.99	1.22	> 0.99
	NDI	0.22	> 0.99	0.01	> 0.99
	ODI	0.72	> 0.99	0.01	> 0.99

		Entire ipsilateral hemisphere			
		Speech perception		Speech comprehension	
		<i>R</i>	<i>P</i> _{FWE}	<i>R</i>	<i>P</i> _{FWE}
Left PT	Surface area	0.01	> 0.99	0.05	> 0.99
	Thickness	-0.01	> 0.99	-0.04	> 0.99
	Myelin content	-0.03	> 0.99	0.00	> 0.99
	NDI	-0.03	> 0.99	0.00	> 0.99
	ODI	0.02	> 0.99	0.01	> 0.99
Right PT	Surface area	0.02	> 0.99	0.00	> 0.99
	Thickness	0.05	> 0.99	0.02	> 0.99
	Myelin content	0.00	> 0.99	0.01	> 0.99
	NDI	0.03	> 0.99	0.04	> 0.99
	ODI	0.04	> 0.99	0.00	> 0.99

Table S10

The correlations of PT structural measures with speech-related functional activation of the entire ipsilateral hemisphere after controlling for the HG gyrification pattern (the second specificity analysis)

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Reviewer #1 (Public Review):

Summary:

Qin and colleagues analysed data from the Human Connectome Project on four right-handed subgroups with different gyrification patterns in Heschl's gyrus. Based on these groups, the authors highlight the structure-function relationship of planum temporale asymmetry in lateralised language processing at the group level and next at the individual level. In particular, the authors propose that especially microstructural asymmetries are related to functional auditory language asymmetries in the planum temporale.

Strengths:

The study is interesting because of an ongoing and long-standing debate about the relationship between structural and functional brain asymmetries, and in particular whether structural brain asymmetries can be seen as markers of functional language brain lateralisation.

In this debate, the relationship between Heschl's gyrus asymmetry and planum temporale asymmetry is rare and therefore valuable here. A large sample size and inter-rater reliability support the findings.

Weaknesses:

In this case of multiple brain measures, it would be important to provide the reader with some sort of effect size (e.g. Cohen's d) to help interpret the results. In addition, the authors highlight the microstructural results in spite of the macrostructural results. However, the macrostructural surface results are also strong. I would suggest either reducing the emphasis on micro vs macrostructural results or adding information to justify the microstructural importance.

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Reviewer #2 (Public Review):

Summary:

The authors assessed the link between structural and functional lateralization in area PT, one of the brain areas that is known to exhibit strong structural lateralization, and which is known to be implicated in speech processing. Importantly, they included the sulcal configuration of Heschl's gyrus (HG), presenting either as a single or duplicated HG, in their analysis. They found several significant associations between microstructural indices and task-based functional lateralization, some of which depended on the sulcal configuration.

Strengths:

A clear strength is the large sample size ($n=907$), an openly available database, and the fact that HG morphology was manually classified in each individual. This allows for robust statistical testing of the effects across morphological categories, which is not often seen in the literature.

Weaknesses:

- Unfortunately, no left-handers were included in the study. It would have been a valuable addition to the literature, to study the effect of handedness on the observed associations, as many previous studies on this topic were not adequately powered. The fact that only right-handers were studied should be pointed out clearly in the introduction or even the abstract.
- The tasks to quantify functional lateralization were not specifically designed to pick up lateralization. In the interest of the sample size, it is understandable that the authors used the available HCP-task-battery results, however, it would have been feasible to access another dataset for validation. A targeted subset of results, concerning for example the relationship between sulcal morphology and task-based functional lateralization, could be re-assessed using other open-access fMRI datasets.
- The study is mainly descriptive and the general discussion of the findings in the larger context of brain lateralization comes a bit short. For example, are the observed effects in line with what we know from other 'language-relevant' areas? What could be the putative mechanisms that give rise to functional lateralization based on the microstructural markers observed? And which mechanisms might be underlying the formation of a duplicated HG?

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